

Module 6. Cluster Analysis and Selected Methods

- ❑ Cluster Analysis: Basic Concepts
- ❑ Partitioning Methods
- ❑ Hierarchical Methods
- ❑ Density-based and Grid-based Methods
- ❑ Advanced Methods 
- ❑ Evaluation of Clustering
- ❑ Summary

Advanced Method: Probabilistic model-based clustering

- Probabilistic model-based clustering
 - Fuzzy clusters
 - Probabilistic model-based clusters
 - Expectation-maximization (EM) algorithm

Fuzzy Set and Fuzzy Cluster

- ❑ Clustering methods discussed so far
 - ❑ Every data object is assigned to exactly one cluster
- ❑ Some applications may need for fuzzy or soft cluster assignment
 - ❑ Ex. An e-game could belong to both entertainment and software
- ❑ Methods: fuzzy clusters and probabilistic model-based clusters
- ❑ Fuzzy cluster: A fuzzy set $S: F_S : X \rightarrow [0, 1]$ (value between 0 and 1)
- ❑ Example: Popularity of cameras is defined as a fuzzy mapping

Camera	Sales (units)
<i>A</i>	50
<i>B</i>	1320
<i>C</i>	860
<i>D</i>	270

$$Pop(o) = \begin{cases} 1, & \text{if } 1,000 \text{ or more units are sold} \\ \frac{i}{1000}, & \text{if } i < 1000 \text{ units are sold} \end{cases}$$

- ❑ Then, $A(0.05)$, $B(1)$, $C(0.86)$, $D(0.27)$

Fuzzy (Soft) Clustering

Example: Let cluster feature be

C1: “digital camera” & “lens”

C2: “computer”

$$M = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ \frac{2}{3} & \frac{1}{3} \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

Review-id	Keywords
R_1	digital camera, lens
R_2	digital camera
R_3	lens
R_4	digital camera, lens, computer
R_5	computer, CPU
R_6	computer, computer game

Fuzzy clustering: k fuzzy clusters $C_1, \dots, C_k$, represented as a partition matrix $M = [w_{ij}]$

P1: For each object o_i and cluster C_j , $0 \leq w_{ij} \leq 1$ (fuzzy set)

P2: For each object o_i , $\sum_{j=1}^k w_{ij} = 1$ (equal participation in clustering)

P3: For each cluster C_j , $0 < \sum_{i=1}^n w_{ij} < n$ (ensure no empty cluster)

Let $c_1, \dots, c_k$ as the centers of k clusters

For each object o_i , the sum of square error SSE is $SSE(o_i) = \sum_{j=1}^k w_{ij}^p \text{dist}(o_i, c_j)^2$, where p is a parameter

For each cluster C_j , the sum of square error $SSE(C_j) = \sum_{i=1}^n w_{ij}^p \text{dist}(o_i, c_j)^2$

For each cluster C, sum of square error $SSE(C) = \sum_{i=1}^n \sum_{j=1}^k w_{ij}^p \text{dist}(o_i, c_j)^2$